

BREMGARTEN, GermanyWMO: **109000**Lat: **47.900N**Lon: **7.617E**Elev: **712**StdP: **14.32**Time Zone: **1.00 (EUC)**Period: **82-92**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.2	17.2	7.0	8.1	12.0	13.3	11.1	19.0	30.5	46.3	27.7	45.8	3.0	200	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.1	89.7	69.7	86.0	67.9	82.7	66.2	70.8	84.6	69.3	82.9	67.5	80.5	6.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	99.1	77.2	64.4	92.9	75.4	62.7	87.5	73.8	35.1	85.5	33.6	82.6	32.4	80.4	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.8	21.5	19.0	7.5	97.5	9.4	2.7	0.8	99.5	-4.7	101.1	-10.0	102.6	-16.7	104.5		
(5)			7.3	74.8	9.5	1.9	0.5	76.2	-5.1	77.3	-10.4	78.4	-17.2	79.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.5	34.7	35.5	44.0	49.7	57.7	63.8	69.3	68.1	62.0	53.0	41.6	37.4	
	DBStd	14.15	9.85	9.32	6.88	6.12	6.64	5.98	5.86	5.19	5.99	6.66	7.73	8.21	
	HDD50	1893	477	411	207	80	12	0	0	0	0	45	265	396	
	HDD65	5328	939	827	652	461	240	94	23	31	129	375	703	854	
	CDD50	2439	3	4	21	70	250	414	600	562	359	137	13	6	
	CDD65	398	0	0	0	0	13	58	158	128	38	3	0	0	
	CDH74	4273	0	1	5	20	209	608	1717	1298	373	42	0	0	
	CDH80	1401	0	0	0	1	32	156	684	437	82	9	0	0	
Wind	WSAvg	6.8	8.4	6.9	8.3	7.3	6.5	7.0	5.5	5.4	6.0	5.9	6.5	7.3	
Precipitation	PrecAvg	37.3	2.3	2.3	2.4	2.8	3.9	3.7	3.7	3.5	3.1	3.3	3.2	3.1	
	PrecMax	48.5	5.2	4.4	8.0	6.0	6.7	6.9	9.2	8.6	6.5	7.8	8.3	5.7	
	PrecMin	28.5	0.2	0.8	0.6	0.2	1.5	1.3	1.5	0.6	1.0	0.0	0.0	0.2	
	PrecStd	5.3	1.4	1.0	1.5	1.6	1.6	1.6	1.6	1.6	1.4	1.5	1.7	1.5	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.6	62.5	73.2	76.9	84.1	89.3	96.6	93.6	86.3	80.4	64.1	59.3	
		MCWB	49.2	51.9	56.0	58.0	64.6	69.9	71.3	68.0	67.0	63.6	54.4	51.4	
	2%	DB	53.5	56.6	63.6	71.6	79.2	84.2	91.4	88.2	82.4	71.4	59.1	56.5	
		MCWB	47.4	48.3	51.3	54.9	61.6	66.5	70.1	67.6	66.2	60.5	52.4	50.0	
	5%	DB	50.2	51.7	58.8	66.5	75.5	80.8	87.4	84.5	78.7	66.2	55.6	53.3	
		MCWB	45.4	46.0	49.2	53.3	60.7	65.5	69.0	66.8	64.6	57.6	50.5	48.0	
	10%	DB	47.8	48.3	55.2	62.5	71.4	77.1	83.8	81.0	74.9	62.5	52.2	49.8	
		MCWB	43.7	44.0	47.2	51.0	58.7	63.4	67.6	65.7	62.9	55.5	47.9	45.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.3	52.0	56.3	59.1	66.3	71.6	74.8	71.2	69.9	63.8	55.9	53.5
			MCDB	56.6	60.4	71.0	74.0	78.9	84.9	90.3	86.8	83.2	76.2	61.2	57.3
2%		WB	47.6	49.2	52.2	56.3	63.2	68.4	72.1	69.4	67.4	61.1	53.2	50.2	
		MCDB	52.8	55.2	61.4	68.3	76.1	81.0	85.9	83.5	79.4	69.1	58.0	55.1	
5%		WB	45.6	46.3	49.9	54.3	61.2	66.4	70.1	68.0	65.4	58.5	51.1	48.3	
		MCDB	49.6	50.8	56.6	64.5	72.4	79.0	83.9	82.0	75.9	64.1	55.6	53.1	
10%		WB	43.4	43.9	48.0	52.2	59.6	64.2	68.3	66.7	63.3	56.5	48.6	45.6	
		MCDB	47.0	47.8	54.9	60.9	69.8	74.8	81.7	79.4	72.5	61.6	52.4	49.2	
Mean Daily Temperature Range		5% DB	MDBR	10.8	14.4	17.1	19.7	20.4	20.2	23.1	23.0	20.8	16.5	12.5	10.5
			MCDBR	13.4	21.2	24.7	29.3	29.4	27.5	30.6	29.7	27.3	22.6	17.5	14.3
	MCWBR		9.8	14.4	14.5	15.7	14.8	13.0	13.1	12.8	13.1	13.3	12.1	10.2	
	5% WB	MCDBR	12.7	18.3	21.2	26.5	25.1	25.3	26.8	26.5	24.6	20.9	16.7	13.2	
		MCWBR	9.8	13.1	13.1	14.4	13.3	12.5	12.3	12.0	12.8	12.7	12.2	10.0	
Clear-Sky Solar Irradiance	taub		0.315	0.341	0.374	0.411	0.421	0.426	0.424	0.419	0.394	0.371	0.336	0.312	
	taud		2.439	2.360	2.297	2.215	2.236	2.260	2.260	2.291	2.336	2.386	2.436	2.437	
	Ebn at Noon		241	258	266	266	268	266	265	261	256	243	229	226	
	Edh at Noon		21	28	35	42	43	42	41	38	33	27	21	19	
All-Sky Solar Radiation	RadAvg		324	574	964	1369	1603	1870	1768	1538	1165	716	372	261	
	RadStd		45	95	148	199	167	165	164	151	115	64	43	49	